Rate-Distortion Analysis of Minimum Excess Risk in Bayesian Learning

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Problem Formulation

- **Generative Model**: $Z^n = \{(X_i, Y_i)\}_{i=1}^n$ and $Z = (X, Y)$.

$$P_{W,Z^n,Z} = P_W \otimes \prod_{i=1}^{n} P_{Z_i|W} \otimes P_{Z|W}$$

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Minimum Excess Risk in Bayesian Learning

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Theorem (Xu & Raginsky 2020)

Consider an arbitrary non-negative bounded function $\ell : \mathcal{Y} \times \mathcal{Y} \to [0, b]$. *We have*

$$\mathrm{MER}_\ell^n \leq \sqrt{\frac{b^2}{2} I(Y; W|Z^n, X)} \leq \sqrt{\frac{b^2}{2n} I(W; Z^n)}.$$

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- Remarks
  - Under mild conditions, $I(Y; W|Z^n) = O(1/n)$ as $n \to \infty$ giving

$$\mathrm{MER}_\ell^n = O(\sqrt{1/n})$$

  - For bounded random variables this can be improved to $O(1/n)$ if the loss is quadratic or the problem is realizable.
  - The lower bound was left as an open problem in [Xu & Raginsky 2020]

An Appetizer for the Rate-Distortion View - Lower Bound

$$W \to Z^n \to \hat{h}(\cdot)$$

- **Goal:** observe Z^n and find $\hat{h}(.)$ which performs well compared to the case where W is known

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- How well we can act, based on a suitable distortion function, if only $I(W; Z^n)$ nats of information about W is received *through* Z^n?
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Question

How efficient is this encoder?
If one is asked to use $R = I(W; Z^n)$ nats to represent W by an intermediate variable Ξ in a way that it is possible to recover a good $\hat{h}$, is it a good idea to just generate n i.i.d. samples from P_Z^W; i.e., use $\Xi = Z^n$?

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Question

Can we limit the information used by the encoder? *Is it possible to have* $I(Z^n; \hat{h}) = I(W; Z^n)$ *and still achieve the optimal* $\hat{h}(.)$.

Rate-Distortion Analysis of MER

- Define the following distortion function:

$$d(w, \hat{h}(\cdot)) = \mathbb{E}^{w}_{XY}\Big[\ell(Y, \hat{h}(X)) - \ell(Y, \psi^{*}_{Y|XW}(w, X))\Big].$$

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- Define the constrained rate-distortion optimization:

$$D_n(R) = \inf_{P^{Z^n}_{\hat{h}}} \mathbb{E}\big[d(W, \hat{h})\big] \quad \text{s.t.} \quad I(W; \hat{h}) \leq R,$$

Theorem

For a given training set size n, for all rates $R \geq I(W; Z^n)$, we have

$$D_n(R) = \mathrm{MER}^{n}_{\ell}.$$

Lower Bound

- One can remove the constraint that $\hat{h}$ is generated only using the samples Z^n:

$$D^L(R) = \inf_{P_{\hat{h}}^W} \mathbb{E}[d(W, \hat{h})] \quad \text{s.t.} \quad I(W; \hat{h}) \leq R.$$

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- For all R and n, we have $D^L(R) \leq D_n(R)$.

Upper Bound

- By adding another constraint that the mutual information between the dataset and the learned model $\hat{h}$ should also be constrained by R:

$$D_n^U(R) = \inf_{P_{\hat{h}}^{Z^n}} \mathbb{E}[d(W, \hat{h})] \quad \text{s.t.} \quad I(Z^n; \hat{h}) \leq R.$$

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- For all R and n, we have $D_n^U(R) \geq D_n(R)$.

Theorem

For any bounded loss function $\ell : \mathcal{Y} \times \mathcal{Y} \to [0, b]$, and for all $n \geq 1$, we have

$$D_n^U(I(W; Z^n)) \leq \sqrt{\frac{b^2}{2} I(W; Y | Z^n, X)} \leq \sqrt{\frac{b^2}{2n} I(W; Z^n)}.$$

Relation Between the Upper and Lower Bounds

The following theorem states the relation between the upper bound $D_n^U(R)$ and the lower bound $D_L(R)$.

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For any bounded loss $\ell : \mathcal{Y} \times \mathcal{Y} \to [0, b]$, we have

$$D_n^U(R) - D^L(R) \leq \sqrt{\frac{b^2}{2} I(W; \hat{h}_R | Z^n)},$$

where $P_{W,\hat{h}_R Z^n} = P_W \otimes P_{\hat{h}_R}^{*W} \otimes P_{Z^n}^W$ and $P_{\hat{h}_R}^{*W}$ is a solution to $D^L(R)$.

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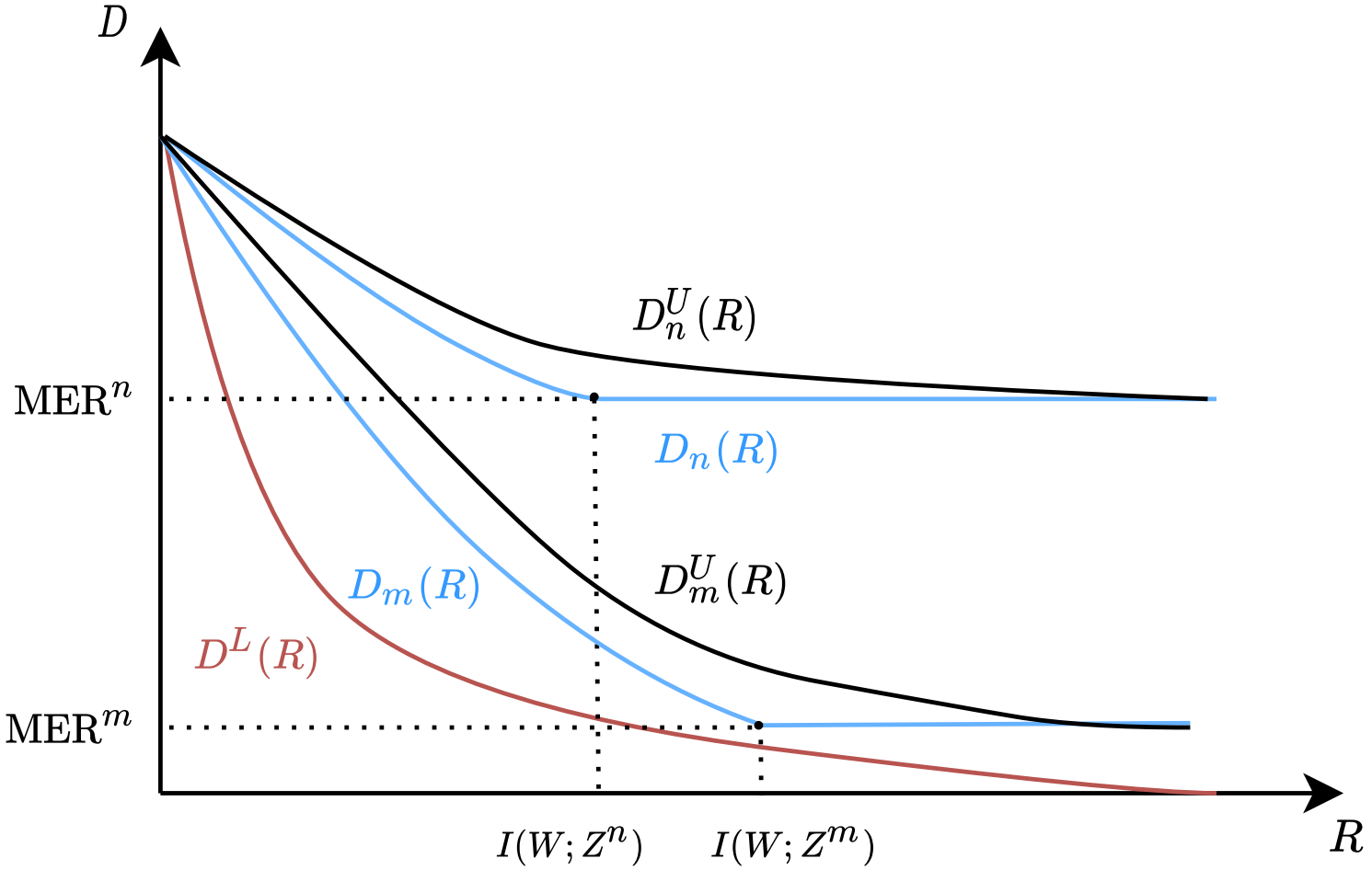
Theorem

Suppose the distortion $d(W, \hat{h})$ can be represented as a distance $d'(h_W^*, \hat{h})$. Let W and W' be two samples independently generated from $P_W^{Z^n}$. If we have $\lim_{n\to\infty} \mathbb{E}[d'(h_W^*, h_{W'}^*)] = 0$, then

$$\forall R \geq 0;\ D^L(R) = \lim_{n\to\infty} D_n(R) = \lim_{n\to\infty} D_n^U(R).$$

Relation Between the Upper and Lower Bounds



Lower Rate

- For quadratic loss we have $d(w, \hat{h}) = ||\psi_{Y|W,X}(w, \cdot) - \hat{h}(\cdot)||_{L^2(P_X)}$.

Theorem

Let $\mathcal{W}$ be a p-dimensional compact subspace of $\mathbb{R}^p$. Under some mild conditions (see Section 6 of the paper):

$$\mathrm{MER}_\ell^n \geq \frac{p}{n} \cdot \frac{\pi}{\left(V_p\, \Gamma(1 + \frac{p}{2})\right)^{\frac{2}{p}}} \exp\Big(- \frac{\mathbb{E} \log |J_Z^W(W)|}{p} \Big),$$

as $n \to \infty$, where V_p is the volume of $\{x \in \mathbb{R}^p : ||x|| \leq 1\}$.

- The bounds are tight for the cases where the upper rate of $O(1/n)$ holds

Linear Regression

Under the conditions that

- The space $\mathcal{W}$ is convex.
- P_W is supported on a compact subspace $\mathcal{W}$ of $\mathbb{R}^p$.
- $Y = W^\top X + \sigma\nu$, where $W \sim P_W$, $X \sim \mathcal{N}(0, \Sigma_X)$, and $\nu \sim \mathcal{N}(0, 1)$.
- Variables W, X, and ν are independent.
- The matrix Σ_X is full-rank.

We have

$$\mathrm{MER}_\ell^n = \Omega(\frac{p}{n}).$$

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Similar results can be derived for Neural Tangent Kernels

$$f(\cdot, w) = f(\cdot, w_0) + \Phi_{w_0}^\top(\cdot)(w - w_0).$$

Thank You!